

Robert Barthell Garay

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EDUCATION

University of Illinois Urbana-Champaign
Bachelor of Science in Aerospace Engineering

3.3 GPA

Expected December 2026

SKILLS

Classes: AE 502 (Advanced Orbital Mechanics), AE 442/443 (Spacecraft Systems Design), ENG 491 (Cubesats)

Languages: English, Spanish (native), Mandarin Chinese (HSK5), Russian, German, French, Hindi

Technical Skills: • FreeFlyer (trajectory propagation, perturbation modeling, maneuver planning, ADCS)

• Python (numerical optimization, adaptive weighting, least squares solvers, data analysis)

UNIVERSITY ORGANIZATIONAL INVOLVEMENT:

Laboratory for Advanced Space Systems at Illinois - Research Assistant **August 2025 - Present**

CubeSail II - Feasibility mission testing deployment and controllability of Solar Sail propulsion

- Coordinated the mission's early concept development stages, literature review, created initial simulation framework
- Refined simulated attitude determination logic for sun-pointing control to maximize solar exposure and efficiency
- Adapted orbital integrator to simulate flexible spacecraft structures under solar radiation pressure
- Created multi-node dynamics soft-body/tether system to model structural deformation in sun synchronous orbit

MonARCH - CubeSat testing a novel electro spray-based hybrid propulsion system

- Programmed long-term orbit propagation simulations producing data about propulsion efficiency vs. external forces
- Modeled coupled ADCS-orbit dynamics accounting for momentum build-up and propulsion-induced disturbances
- Built and implemented custom models for counteracting misaligned thrust and maintaining attitude stability
- Achieved sustained reaction wheel momentum containment using optimized torque rod desaturation strategy

Illinois Space Society - Media Team, 3D Animator, Undergraduate Design Member **September 2023-April 2025**

- Analyzed design of an orbital laser satellite communications network around Moon using FreeFlyer
- Digitally reconstructed the Lunar South Pole in Unreal Engine and created 3D animations of proposal designs
- Produced video recaps and written reports, directed advertising shown to hundreds of prospective/new members

Chinese Language Association for International Development - Board Member **January 2024 - Present**

- Led discussion and conversation tables for Chinese language learners, contributed to cultural immersion programs

ACADEMIC PROJECTS:

Advanced Orbital Mechanics Semester Project **February 2026 - April 2026**

- Constructed n-body propagator to design multiple gravity-assist trajectory ($E\Delta vEJU$) for Uranus orbiter mission
- Implemented shooting method with least squares solver which achieved 40% reduction in needed Δv for probe
- Expanded results from published NASA studies, gained experience with nonlinear trajectory optimization

Spacecraft Systems Design Proposal **January 2026 - May 2026**

- Co-wrote research paper proposing twin spacecraft solution for AIAA Solar Weather Monitoring design competition
- Designed trajectories for Mars-Sun L1 probe and inclined relay probe, ensuring constant Earth-Mars line of sight

WORK EXPERIENCE:

University of Illinois - AE 402 Course Assistant **August 2026 - Present**

- Produced and edited hyperrealistic spacecraft trajectory render multimedia to supplement class presentations
- Developed and maintained interactive orbital mechanics assignments with randomized numerical scenarios
- Rebuilt class assignments on new digital learning platform with integrated Python evaluation and instant grading

National Aeronautics and Space Administration - Software Development Internship **June 2024 - August 2024**

- Developed high-fidelity training content and environments for NASA simulator of Lunar/Martian surface
- Created 3D models of EVA tools, habitats, and equipment which will be used to train future astronauts
- Collaborated with software engineers to implement interactive physics-based systems and scenarios